

# Mika Be

## APPLIED AI | SOFTWARE ENGINEERING

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## EDUCATION

**Rutgers University, New Brunswick** | B.S. Electrical and Computer Engineering, CS minor Expected May 2028

**GPA** 3.838/4.000 | Dean's List, all four completed semesters

**Honors** 2nd Place Best Full Paper, IEEE ISEC 2026 | IEEE-HKN honor society | Bergen County Academies CS Award (1 of 3)

**Coursework** Data Structures; Digital Logic Design + Lab; Principles of EE I & II + Labs; Discrete Structures; Differential Equations

**In progress** Computer Architecture & Assembly + Lab; Linear Systems & Signals; Electronic Devices + Lab; Discrete Structures II

## RESEARCH

**Kuiper TTS, Rutgers ECE** | Undergraduate Research Lead (PI: Prof. Dov Kruger) Sep 2025-Present

*Python* | *PyTorch Lightning* | *Piper/VITS* | *ONNX*

- Led an 18-member team building offline text-to-speech to read scientific papers for blind and low-vision users
- Co-authored the 2nd Place Best Full Paper at IEEE ISEC 2026; designed its rules for reading equations aloud, tested on 46 equations from a 1971 antenna paper (“ $d q$  over  $d t$ ” became “the derivative of  $q$  with respect to  $t$ ”)
- Built a GPU training pipeline and trained custom voices on 886 self-recorded clips; fixed a pitch bug in augmentation
- Created audio-cleanup, augmentation and phoneme-coverage tools the team ported to C++; wrote the voice-training guide

## EXPERIENCE

**this.is** | Lead Software Developer (Internship) Jul 2025-Aug 2026

*React* | *TypeScript* | *Firebase* | *Cloud Functions* | *Google Places API*

- Developed and deployed a startup's social place-discovery web app as primary engineer; mentored and managed 2 interns
- Built a Google Places recommendation engine favoring local businesses, with novelty tracking and geographic diversity
- Led a product pivot and built v2 as sole developer: a group planner that turns saved places into up to 3 suggestions

**Schieber Research** | Software Engineering Intern Sep 2023-Jun 2024

- Analyzed competitors and wrote product documentation for this.is, a startup client building a discovery app
- Designed and built the app's first 3 prototypes in Node.js, React and Appwrite

**Starbucks** | Barista (part-time, alongside full-time coursework) Jul 2023-Jun 2025

## PROJECTS

**Room of Days** | Creator and Solo Developer, RPG-style habit app for iOS Jun 2026-Present

*Flutter* | *Dart* | *Swift (WidgetKit)* | *Firebase* | *TypeScript Cloud Functions* | *Codemagic CI*

- Shipped to the App Store in Aug 2026 as sole designer and developer; 10+ active users, 5.0 stars (5 ratings)
- Designed the XP system: 6 life stats, diminishing returns on repeats, difficulty ladders; no penalty for missed days
- Deployed a Firebase social backend: room sharing, an opt-in directory, server-side moderation and App Attest checks

**Room Notes** | Creator and Solo Developer, cross-device note-taking app (private beta) Aug 2026-Present

*Flutter* | *Dart* | *Objective-C++* | *Drift/SQLite* | *Firestore* | *Cloud Functions*

- Built a local-first notes app where iPad (Apple Pencil), phone (board photos) and laptop edit the same pages
- Traced a 1.5 s pen freeze after writing pauses to sync-echo handling and removed it, verified in a sync simulation
- Patched Flutter's iOS engine to deliver coalesced (between-frame) Apple Pencil input; shipped it via CI to TestFlight

**Knowsy Breadboard Lab, Rutgers ECE** | Team Lead, 3-person team (1 of 7 module leads) May 2026-Present

*JavaScript* | *SVG* | *Verilog*

- Created a browser breadboard simulator for Digital Logic and EE II lab prep, matching real lab boards and 74LS pinouts
- Developed live breadboard-to-schematic rendering and debugging tools: logic probe, voltmeter and wiring-fault warnings
- Adapted course materials into 8 guided labs (90 auto-checked steps); built a nodal-analysis solver and Verilog export

**Bio (Discord bot)** | Creator and Developer Sep 2025-Present

*Node.js* | *SQLite* | *Railway*

- Launched a Discord bot that connects members with shared interests through opt-in profiles; live in 2 servers

## TECHNICAL SKILLS

**Programming** Python, TypeScript, JavaScript, Dart, C/C++, Java, SQL, Swift, Verilog

**AI/ML** PyTorch Lightning, ONNX, Piper/VITS TTS training, speech data prep (VAD, LUFS), AI coding agents (Claude Code, Codex)

**Software** Flutter, React, Node.js, Firebase (Firestore, Cloud Functions, App Check), Drift/SQLite, Git, CI/CD (Codemagic)

**Hardware and EE tools** Vivado, LTspice, MATLAB/Simulink, oscilloscopes, logic probes, soldering, Onshape CAD, 3D printing

**Languages** English (fluent), Hebrew (native), Spanish (conversational)